ROUND 2 · THE BUILD
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October 2026

Plumbline

A pre-publication checkpoint that tests copy, claims and measurable visual rules against the organisation's own rules, facts and approved wording. It shows the evidence for each issue, says what it did not check, and learns which guidelines are unclear. People make every decision.

THE PROBLEM	Content comes from employees, agencies, influencers and AI tools faster than a brand team can review it. Wrong claims, stale facts, off-brand tone and visuals, and accessibility gaps reach the public, and the same mistakes repeat because corrections never feed back.
WHO HAS IT	Small brand teams in mid-size organisations (first: Indian D2C personal care) who govern content from dozens of creators; compliance reviewers who sign off claims; and creators and agencies, who wait days for vague feedback.
HOW I SOLVE IT	Turn the brand book into **testable rules** and a **live fact ledger** with owners, expiry dates and **approved wording**. Code checks first, AI only for judgement. **Every finding cites its source, every report lists what was not checked, and claims are fixed only with approved wording.** Legal and accessibility rules block from day one; brand rules earn the right to block.
WHAT I BUILD, IN ORDER	① Rulebook, fact ledger and approved wording → ② text checks with cited findings → ③ fixes: approved-wording swaps and the intent lock → ④ review workflow and guideline health v1 ROUND 2 · 40 H → ⑤ integrations (Slack, link, publish gate) → ⑥ deeper visual checks → ⑦ claim recall and rule maturity → ⑧ evaluation and hardening ROUND 3 · 78 H + BUFFER

The three improvements I commit to deliver (v1 by Round 2; measured by Round 3)

1 Evidence or silence, and silence is never clearance

Every finding cites a rule or fact, quotes the text and passes a second-model support check. Every report lists what was not checked.

Target: ≥ 95% citation correctness (human-checked); recall ≥ 90% on seeded claims, ≥ 80% on public ASCI decisions

2 Fixes that keep intent and never invent claims

Claims are fixed only with approved wording from the ledger; other copy gets minimal rewrites that pass five intent checks.

Target: 0 invented claims in a 200-fix audit; ≥ 70% of fixes accepted in user tests

3 Guidelines get clearer; rules earn the right to block

Override and disagreement rates per rule. Brand rules stay advisory until reviewers agree with them; unclear rules get a drafted rewrite.

Target: top-5 "unclear rules" includes ≥ 2 of the 3 rules testers name as most confusing, before seeing it

Where each Round 1 requirement is answered

REQUIRED ITEM	SECTION	JUDGING CRITERION
Problem definition and evidence that the problem is real	1	Understanding (25)
Target users, their needs and the organisational setting	2	Understanding (25)
Research: desk research, an audit with a blind AI second rater, a labelled synthetic stress test, interview plan	3	Understanding · Originality
Proposed solution, value proposition and end-to-end workflow	4	Solution (25) · Originality (15)
Planned features, technology stack, integrations and data sources	5, 6	Technical plan (20)
Responsible design: privacy, security and human oversight	7	Solution · Technical plan
Development plan with buffered hours, role allocation and expected impact	8, 9, 10	Feasibility and scope (15)

ROUND 2 · THE BUILD

R2 Round 2 progress

Plumbline is live and works end to end from a public link: paste or upload content, get cited findings and a "not checked" list, accept approved-wording fixes, pre-check privately, submit, review with routed approvals, and verify the audit trail. Round 1's plan follows unchanged after this section.

> **Live app:** getplumbline.xyz (pick a role, no sign-up) · **How I know it works:** /about · **Repository:** github.com/vaibhav375/plumbline · **Status:** /status. Demo runs on free model tiers using fictional data only; production uses paid no-training tiers or self-hosted Qwen.

What was built (all live)

AREA	WHAT WORKS	ROUND 1
Checking	Paste, DOCX, PDF or image (OCR). 28 code rules run first; AI only for judgement and claims. A finding is shown only if its rule or fact exists and its quote is in the text; a second model checks AI findings. Every report lists what wasn't checked.	§4.1-2, §4.3, §6.2
Fixing	Claims fixed only with approved wording (with required footnotes); other rewrites pass five intent checks. Accept, edit, dismiss with a reason, or ask a reviewer; re-check with a diff.	§4.1-4, §6.2 stage 7
Privacy	Names and contacts masked before any model call; confidential and embargoed facts matched in code only; an egress log with no field for content; agencies never see confidential facts.	§4.5, §7
Review and audit	Private pre-checks; a risk-ordered queue; claims routed to compliance; a different second approver enforced in the database; an insert-only, hash-chained audit log that verifies in the app and exports a CSV anyone can re-check.	§4.5
Governance	Rulebook with a linter and preview-first CSV import (each function edits only its own rules); fact ledger with expiry and approved wording; guideline health with Wilson lower bounds ("ready to block", "may be unclear").	§4.3, §4.6
Accessibility floor	Image text contrast (4.5:1, 3:1 large) and missing alt text block; findings are boxed on the image; Plumbline never writes alt text.	§4.3, §5

How I know it works

DEV SET, 40 POSTS	PLUMBLINE	BASELINE
Problems found (recall)	**31 / 33 · 94%** 95% CI 80–98%	18 / 33 · 55% 38–70%
Findings that were right	**31 / 39 · 79%** 64–89%	18 / 37 · 49% 33–64%
Clean posts with a false alarm	**1 / 10**	4 / 10
Characters sent to models	243k	634k

25 labelled posts (33 problems), 10 clean controls, 5 prompt-injection copies; written for this project, labels frozen before any model run. A finding counts only with the right rule or fact and the right words. Baseline: one model call per post with the whole brand book, no validators.

A negative result. "Citations that check out" was 100% for both systems: the baseline cites real IDs and real quotes, just the wrong ones. Precision is what separates them. Two of Plumbline's 8 false alarms are plain errors.

Tests. 251 checker unit tests, 17 database tests (row-level security, audit chain, second approver), 18 end-to-end tests run against the live site.

Speed on free tiers. Text checks 1.8–3.6 s; image checks about 6 s, over the 5 s target because OCR alone takes ~3.8 s on the shared CPU. The first check after a restart fell from ~20 s to 4.6 s by loading the privacy model at startup.

Build log. Every one of the 24 tasks was evaluated against real runs, and each assumption that failed is recorded with what replaced it (*docs/build-log.md*).

Round 1 commitments for Round 2, and where each stands

ROUND 1 SAID	ROUND 2
v1 of all three improvements	MET Evidence or silence; fixes from approved wording with the intent lock (10/10 live posts kept intent, 0 invented claims); guideline health v1
§6.7 recall ≥ 80% on seeded claim issues (held-out set)	PARTLY 13 / 14 claim and fact labels (93%) on the builder's dev set; no held-out set yet

§6.7 ≥ 90/100 citations correct, checked by a person	PARTLY 52 / 52 valid, checked automatically only
§3.5 interviews and survey; §3.2 human raters; §6.7 held-out labellers; §4.6 tester back-test; §8/§10 user tests with 6 people	NOT RUN They need recruited participants, which a one-person team can't run in a hackathon. The evidence plan changed instead (below). No AI-simulated persona stands in for them
Second approver and publish gate in Round 3	AHEAD The second approver shipped in Round 2, enforced in the database

How the evidence plan changed, and why

Round 1 planned to prove Plumbline with recruited people. Plumbline is built by one person, and recruiting and running studies with outside participants doesn't fit a hackathon's build windows. So the plan moved from panels the builder would recruit to evidence anyone can verify: the competition asks that everything be openable, runnable and checkable, and a private panel is none of those. Each question Round 1 asked is kept, and answered with a source independent of the builder or a check one person can do and anyone can re-run.

ROUND 1 PROMISED	INSTEAD	WHAT IT PROVES
Test set labelled by two recruited marketers	Public ASCI (India) and UK ASA rulings on beauty and personal-care ads; the regulator's verdict is the label (Round 3)	Catches what regulators act on, on labels independent of the builder
100 citations checked by a person	Checked by the builder against the rule or fact each cites (Round 3)	Every finding points at real evidence
User tests: ≥ 70% of fixes accepted	A 200-fix audit (0 invented claims) and automatic intent checks (Round 3)	Fixes are safe and keep the message
Testers name confusing rules; testers' 60-post back-test	Recorded decisions replayed, with regulator rulings as real upheld or not-upheld decisions (Round 3)	Clarifications reduce disagreement, on real decisions
Interviews and a survey	Published surveys, labelled as desk research (below)	The problem exists at scale; not primary research

What this doesn't show: regulator rulings test whether Plumbline finds the right problems, not whether writers enjoy using it. No user study is claimed, and none is simulated with AI personas. Every number can be re-run from the public repository.

Desk research in place of interviews (labelled as such)

HYPOTHESIS	VERDICT	STRONGEST PUBLIC EVIDENCE, AND ITS LIMIT
H1 Most review time goes on mechanical issues	PARTLY	ASCI, Feb 2025: 69 of India's top 100 influencers broke disclosure rules (56.8% no label, 43.2% buried in hashtags), and 59 of 69 fixed it without contest. Doesn't measure reviewer time.
H2 Creators prefer an editable suggested fix	WEAK	GitHub, May–Jul 2024: suggested fixes resolved code alerts in a median 28 min vs 1.5 h. Vendor data, another profession, speed not preference.
H3 Teams interpret some rules differently	PARTLY	Lucidpress 2019: 81% of 200+ organisations deal with off-brand content; ASCI: 43.2% of violations had a disclosure, placed wrongly. Not the "name three rules" test.
H4 Claim evidence lives in email and spreadsheets	NO EVIDENCE	None found; pharma uses dedicated claims libraries, so likely false there.
H5 Pay per organisation, not per seat	NO EVIDENCE	Comparable tools price per user, flat or by usage; says nothing about buyer preference.
H6 The synthetic panel's conditions matter	PARTLY	Salesforce: 41% of CMOs name data exposure their top AI concern; 31% of 1,000+ marketers rank accuracy first. Supports code-only confidential facts and evidence or silence.

Changes from Round 1

ROUND 1 SAID	ROUND 2 DOES	WHY
Anthropic Claude API (Haiku extracts, Opus judges)	Open-weight Qwen3.8-27B and GPT-OSS on Groq; Gemini Flash fallback	₹0 budget; open weights enable self-hosting
Paid model API with redacted brand data	Free tiers, fictional data only	Free tiers may use inputs

Supabase Pro; a paid 2 GB worker	Free tiers, kept awake by a database cron job every 10 min	₹0; images run slower on the free CPU
Audit chain head e-mailed daily	Chain verified in the app and exportable	External anchor later; not a Round 3 commitment
Postgres full-text search for facts	In-memory fuzzy matching	40 facts; full-text search when the ledger grows
Back-test on reviewer decisions	Synthetic history, labelled in the app	No real decisions yet
Creators see org-wide insights; any owner edits any rule	Own work only; each function edits only its own rules (both enforced in the database)	Round 1's own permissions table (§4.5), caught during the build

Three improvements for Round 3 (each one deliverable and measurable by one person)

1 Claim watch, tested on regulator rulings

When a fact expires or changes, every published post that used it gets an update-or-withdraw task with the replacement wording; published URLs are re-checked against the approved version.

Held-out ASCI and UK ASA rulings: recall ≥ 80% on upheld claims; ≥ 95% of 100 citations correct (builder-checked); repeated on a second brand loaded by CSV only

2 Each brand's own checks, in chat and before publishing

A brand links its ASI:One address to its Plumbline organisation with a one-time code, so chat checks use its own rules and facts; a publish-gate API tells scheduling and CMS tools "not yet" while anything blocks.

200-fix audit with 0 invented claims; automated tests: the gate refuses 100% of blocked versions, no brand sees another's facts

3 Self-learning guidelines

Brand rules graduate to blocking automatically at a Wilson lower bound ≥ 0.80, with an audit entry and undo; Plumbline drafts clearer wording for "may be unclear" rules from recorded decisions.

Replayed decisions, incl. regulator rulings: accepted clarifications agree more often than the old wording

AI-use disclosure. Built with Claude Code (Anthropic) as a coding assistant, under my direction: I chose the design and every trade-off, reviewed the changes, ran and judged the evaluations, and am the sole author of every commit. The dev set and the fictional Aurelle data were drafted with the assistant. At runtime Plumbline uses open-weight models on Groq with Gemini and ASI:One as fallbacks, on fictional data only.

UNDERSTANDING THE PROBLEM

1 The problem, and evidence that it is real

Brand governance used to be a design question, such as whether the logo sat on the right background. It is now a question of volume, accuracy and legal liability.

1.1 More content, from more hands, including machines

71%

of 1,600+ marketers expect content demand to grow at least fivefold by 2027; 62% say it already grew 5× or more in the prior two years (Adobe, 2025). [1]

>70%

of 200+ CMOs surveyed by BCG (2024) were concerned about generative AI's impact on creativity and brand voice. In Gartner's survey of 418 marketers, 77% explore generative AI but only 44% report significant benefits. [2][3]

77%

of professionals saw off-brand content produced at their company; only 31% said brand guidelines were enforced consistently (Lucidpress survey, n = 452, 2021, as reported by Contently; dated, but the most recent figure I found). [4]

1.2 Each failure in the problem statement shows up in public data

FAILURE NAMED IN THE BRIEF	EVIDENCE, WITH ITS LIMITS
Incorrect and unsupported claims	India's Advertising Standards Council scrutinised 7,199 ads in FY2024-25 and found that **98% required modification**, as did 98% of the 1,015 influencer ads it investigated. *This is a targeted sample: ASCI selects ads through complaints and its own monitoring, and offshore betting made up 43% of cases.* [5] A 2020 EU screening of websites judged **42% of green claims** possibly exaggerated, false or deceptive, and in 59% the trader gave no easily accessible evidence. *Also targeted, not a base rate.* [6]
Outdated or contradictory information	Illustrative case: in *Moffatt v. Air Canada* (2024 BCCRT 149) the airline was held liable after its website chatbot contradicted its own bereavement-fare policy page; the tribunal held the company responsible for all information on its site. [7]
AI-generated errors	Illustrative case: Deloitte Australia agreed to partially refund a roughly A$440,000 government report found to contain fabricated references and a made-up court quote; the revised version disclosed its use of generative AI (October 2025). [8]
Accessibility	**95.9%** of the top one million home pages have detectable WCAG 2 failures; low-contrast text appears on 83.9% and missing alt text on 53.1% (WebAIM Million, 2026, automated testing only). [9]
Slow, repetitive review	The average number of revisions before approval rose from three rounds in 2007 to five in 2025 (IPA, WFA and partners, *The BetterIdeas Project*). [10]

1.3 Why now: in India, a wrong claim now costs the brand and the endorser

Misleading ads and endorsements

The CCPA's guidelines, notified on 9 June 2022 under the Consumer Protection Act, 2019, set conditions for non-misleading ads and disclaimers and make endorsers do due diligence. Penalties reach ₹10 lakh, or ₹50 lakh for repeat violations, for manufacturers and endorsers. [26]

Greenwashing

The CCPA's Guidelines for Prevention and Regulation of Greenwashing or Misleading Environmental Claims (15 October 2024) require terms such as "eco-friendly", "natural" and "sustainable" to be qualified and substantiated. [13]

Influencer disclosure

ASCI's influencer guidelines (in force since 2021) require an upfront label from an approved list (Advertisement, Ad, Sponsored, Collaboration, Partnership, Employee, Free gift) whenever there is a material connection, superimposed on image-only posts. In FY2024-25, 98% of the 1,015 influencer ads ASCI investigated needed changes. [5][27]

Beyond India (roadmap regimes)

The EU bans generic green claims from 27 September 2026 [11]; the UK CMA can fine up to 10% of global turnover since April 2025 [12]; the European Accessibility Act applies from 28 June 2025. [14]

1.4 Root causes (my analysis)

1. **Guidelines live in PDFs, not where content is made.** Agencies, influencers and AI tools never see them at the moment of writing.
2. **Facts have no single owner, expiry date or approved wording.** Prices, study results and launch dates change; old copy keeps being reused, and nobody can find where an outdated claim is still live.
3. **Review is manual, late and sequential.** Mechanical issues use up reviewer time that should go to judgement calls.
4. **Checks look for what is wrong, not for what is missing.** Required disclaimers, risk warnings and paid-partnership labels are verified by memory.
5. **Feedback does not build up, and many guidelines cannot be tested as written.** In the style guide I audited (§3.2), 15–20% of rules have no threshold or example, rising to more than half of its voice-and-tone rules.

> **Problem, in one sentence:** organisations cannot check, at the speed content is now produced, that what they publish is accurate, on-brand and accessible, and they cannot learn from the corrections they already make.

UNDERSTANDING THE PROBLEM

2 Who has the problem, and where they work

2.1 Organisational setting

Mid-size organisations, roughly 200 to 5,000 employees, where a small central brand or marketing-ops team governs content created by many others: regional and product marketers, sales and HR, external agencies and freelancers, influencers, and increasingly AI writing and image tools. The pain is sharpest where claims carry legal risk: consumer goods and D2C (beauty, food, wellness), fintech, health and education.

> **Beachhead for the build:** an Indian D2C personal-care brand, under one regime (ASCI and the CCPA; EU rules are on the roadmap). It makes many claims, relies on agencies and influencers, and publishes on Instagram (organic and influencer), Meta Ads, its Shopify store and email. The fictional brand **Aurelle Skin** is used throughout this plan and in the demo. Team sizes and workflows are working assumptions that the interviews in §3.5 will test.

2.2 Users, needs and what Plumbline gives each of them

ROLE	WHO	PAIN TODAY	WHAT THEY NEED	WHAT PLUMBLINE GIVES THEM
Creator	Marketer, social executive, designer	Waits days for review; vague "off-brand" comments; checks mechanics by hand late at night	Fast, specific feedback; ship on time; keep their idea	A **private pre-check** in about a minute, each finding with the exact rule and a ready fix; results in Slack
External creator	Agency, freelancer, influencer	Works from an outdated PDF; one portal per client; fears blame	Current rules without logins or exposure	A **no-login check link** per brief; private pre-checks; the same insight views about its work that the client sees
Brand reviewer	Brand manager, design lead	Reviews commas, casing and logo spacing instead of decisions	Review less, decide better, stay consistent	Mechanical issues cleared before review; a risk-sorted queue; approved exceptions that stick
Compliance reviewer	Legal, regulatory or medical-claims reviewer	Claims buried in copy; evidence in email; edits after sign-off	Every claim substantiated; an audit trail; nothing invented	Claim-by-claim evidence; fixes only from approved wording; tamper-evident, exportable log
Brand owner	Head of brand (the admin)	Cannot tell whether people are careless or the guideline is unclear	Guidelines that are followed and understood	Guideline health, rule maturity, and recall tasks when a fact changes

2.3 Jobs to be done (my hypotheses, tested in §3.5)

- **Creator:** when I finish a draft, I want to know what will be flagged before a reviewer sees it, so I don't lose a round. *Never:* a public scoreboard of drafts.
- **Compliance:** when a claim reaches me, I want its evidence beside it, so I can approve in minutes. *Never:* a tool that invents claim wording.
- **Brand owner:** when a mistake keeps recurring, I want to know whether people are careless or my guideline is unclear. *Never:* confidential facts leaking.

3 Research and supporting evidence

3.1 Desk research

I reviewed 27 sources: industry surveys (Adobe, Gartner, BCG, Lucidpress, IPA), regulators and enforcement data (ASCI, CCPA, European Commission, CMA, EU directives), a tribunal decision, accessibility data (WebAIM) and the public product material of ten existing tools (Appendix A). The search used AI assistance; every figure was cross-checked against its cited source or published coverage of it, and the limits of each figure are stated where it is used.

3.2 Original observation: how testable is a real style guide?

Method. I took the Mailchimp Content Style Guide (public, CC BY-NC 4.0), a widely cited mature guide, and listed every directive rule in five sections: 114 rules. Each was given one class, and a second rater classified all 114 blind. Appendix B lists every rule and both ratings.

A · CODE-CHECKABLE

A pattern, word list, count or markup check decides it.

e.g. "Use the % symbol instead of spelling out percent"

B · NEEDS UNDERSTANDING

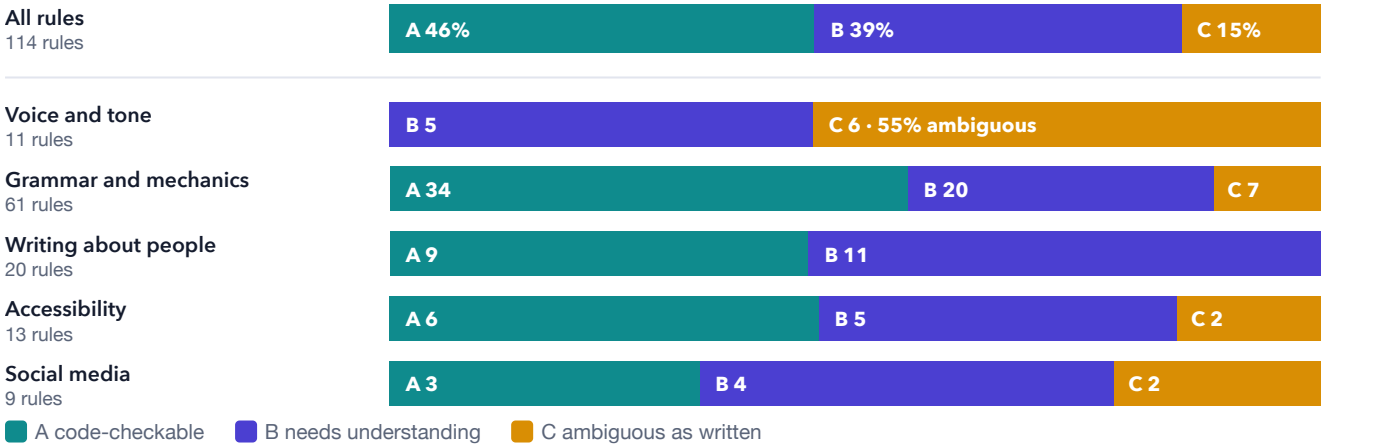
Needs language or visual understanding, but has a clear criterion.

e.g. "Use positive language rather than negative language"

C · AMBIGUOUS AS WRITTEN

No threshold or example; careful reviewers could disagree.

e.g. "Use emoji infrequently and deliberately"



An AI baseline, to be compared with human rating. Both raters so far are AI-assisted (the author working with Claude, and Claude run blind). I use this AI rating deliberately as a baseline: in Round 2, human raters will classify the same rules blind, and I will report how closely AI and human judgement agree. That comparison matters to the product itself, because Plumbline relies on AI to sort rules into these classes. Until then, these figures show the scheme is applied consistently by AI, not that humans agree. The blind AI rater agreed with the first rater on 94 of 114 rules (82.5%, Cohen's $\kappa = 0.72$). It found more ambiguity, not less: 23 rules (20%) ambiguous overall and 9 of 11 voice-and-tone rules; only one rule left class C (the contrast rule, which WCAG makes measurable). **Notable details:** five separate rules say "sparingly" or "rarely" (emoji, exclamation points, ellipses, semicolons, hashtags) without a number, and the accessibility section says "aim for high contrast" without the WCAG 4.5:1 ratio.

FINDING	DESIGN DECISION IT DRIVES
In this guide, nearly half of the rules (46–48%) need no AI at all	**Code first**, and a **code-only mode** for organisations that cannot send content to any AI provider
A third or more (32–39%) need judgement	**AI with a cited rule, a quoted span and a support check**, or the finding is not shown
15–20% give no threshold or example, so reviewers can disagree	A **rule linter** flags them on import (Round 2); they do not run until the owner adds a threshold or examples
Brand voice is the least testable part (55% ambiguous)	Disagreement will cluster here, so this is where **guideline health** matters most

Limitations: one guide, which is grammar-heavy, so the shares will differ for claims-heavy brand books; in Round 2 I will add human raters, with a worked example of each class in their instructions, and repeat the audit on a D2C brand book and one more public guide.

3.3 Existing tools: what is already solved, and the gap

My wedge in one line: the claim discipline pharma and finance already pay for, at a mid-size price, for consumer brands whose content comes from agencies, influencers and AI. Much of this space is served. Writing-governance tools (Markup AI, Writer, Grammarly Business) check style and flag claims; AI proofing and DAM agents check custom rules and logos; regulated industries already pay for **claims libraries with "where used" tracking** (pharma) and **AI compliance review** (finance). Plumbline's position is to bring that claim discipline to **mid-size consumer brands** whose content comes from agencies, influencers and AI tools, and to add three things I did not find advertised by any tool I reviewed: **rules that earn the right to block, reports that state what was not checked, and guidelines diagnosed by reviewer decisions.** It can sit in front of an existing proofing tool rather than replace it.

CAPABILITY	WRITING GOVERNANCE Markup AI, Writer, Grammarly [16][20]	AI PROOFING / DAM Filestage, Bynder [21] [22]	REGULATED COMPLIANCE Veeva, Saifr, Red Marker [23][24] [25]	GENERATION SUITES Jasper, Adobe [17][18]	PLUMBLINE planned
Rule, tone and terminology checks on text	●	●	●	●	●
Visual brand checks (logos, colours)	–	●	◐	●	●
Flags claims that need substantiation	●	◐	●	–	●
Checks that required content is present	–	●	●	–	●
Claims library with "where used" tracking	–	–	● pharma	–	●
Review and approval workflow	◐	●	●	◐	●
Claim fixes restricted to approved wording	–	–	◐	–	●
Report states what was not checked; no pass score	–	–	–	–	●
Rules earn the right to block (shadow mode, per-rule agreement)	–	–	–	–	●
Guideline health from reviewer decisions	◐	–	–	–	●
Code-only mode with no external AI calls	◐	–	–	–	●

From vendors' public product pages, help articles and press releases, October 2026. ● advertised by at least one listed product · ◐ partial (e.g. via custom rules; suggested wording that is model-generated; issue analytics per guideline; a rules engine without generative AI) · – not found in the public material I reviewed, which is not a claim that it does not exist. Plumbline's column is planned, not built.

3.4 Synthetic stress test (design critique, not user research)

Before real interviews, I stress-tested the concept with eight AI-simulated personas, one per role, who answered the interview guide before seeing the concept and then made a pilot decision. Each answered "yes, if…", and the conditions shaped several design choices (approved-wording swaps, the coverage statement, private pre-checks, the accessibility floor), and each is now hypothesis H6 for real interviews to confirm or overturn. Nothing from the panel is used as evidence of demand; details are in Appendix E.

3.5 Primary validation plan (before the window; results reported in Round 2)

Status: not yet conducted at submission. **Plan:** before the window opens (rule 2 allows research), 6 to 8 interviews (brand managers, agency writers, at least one compliance reviewer) and a 10-question survey (target 30+ responses), using Appendix C; continued during the window if it is longer than a few days. Each hypothesis states what I would change if it is false.

#	HYPOTHESIS	IF FALSE, I WILL…
H1	Reviewers spend most review time on mechanical issues.	Shift effort from code checks to judgement quality and routing.
H2	Creators prefer an editable suggested fix to a list of violations.	Make suggestions opt-in and lead with explanations.

H3	Brand owners can name at least three rules their teams interpret differently.	Give guideline health a lower priority than insights by source.
H4	Claim evidence lives in email and spreadsheets, not a system.	Integrate with their system instead of building ledger screens.
H5	Brand owners would pay per organisation from the marketing budget, not per seat.	Test per-seat or agency-paid pricing instead.
H6	The conditions raised by the synthetic panel (Appendix E) matter to real reviewers.	Drop the design choices real users do not confirm.

THE PROPOSED SOLUTION

4 Plumbline: the solution

Value proposition. Plumbline lets any creator (an employee, an agency, an influencer or an AI tool) publish content the brand team would approve. It checks content against the organisation's own rules, facts and approved wording, shows the evidence for every issue, says what it did not check, and fixes problems without inventing claims or losing what the creator meant.

4.1 Six principles that balance automation with human control

1 · Code first, AI second

Plain code decides whatever code can decide (nearly half the rules in the guide I audited). A code-only mode makes no external calls.

2 · Evidence or silence; silence is not clearance

A finding without a valid citation and quote is never shown, and every report lists what was not checked. No pass badge or score.

3 · Suggest, never silently change

One-click fixes still need a click. Nothing is auto-published, and accessibility and claim fixes are never applied automatically.

4 · Never invent a claim

Regulated claims are fixed only with approved wording from the ledger. Other rewrites must pass five intent checks.

5 · Two tiers of rules

Legal and accessibility rules block from go-live, after a setup back-test. Brand rules start advisory and earn the right to block once reviewers reliably agree with them.

6 · Confidential facts stay out of prompts

Confidential facts never enter a prompt. Draft text that matches them is masked, with the catch rate measured; code-only mode closes the gap. Agencies never see a match.

4.2 End-to-end workflow

BRAND OWNER 1
Set up
Import rules, facts with approved wording, terms and assets. Back-test rules on past content.

CREATOR 2
Pre-check
Paste, upload, or (Round 3) Slack and a no-login link. Optional intent. Private until submitted.

SYSTEM · SECONDS 3
Check
Code checks → AI judgement → citation and support check → findings plus a "not checked" list.

CREATOR 4
Fix
Approved-wording swaps for claims; minimal rewrites for the rest. Accept, edit or dismiss with a reason.

REVIEWERS 5
Review
Routed by risk: claims to compliance, visuals to design. Comment, request changes, decide.

SYSTEM 6
Approval record
Bound to a version hash; records which facts each asset uses. Gates any CMS that can ask.

BRAND OWNER 7
Learn
Fact changes raise recall tasks for every asset using them; guideline health drafts clearer rules.

Loop: step 7 feeds step 1, so clearer rules and fresher facts give fewer false flags next time

4.3 The eight building blocks

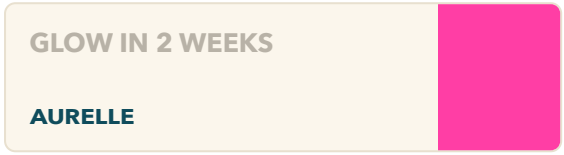
BLOCK	WHAT IT DOES
A · Rulebook	Rules with ID, category, class (A, B or C), severity, threshold, examples, source page, owning function and **tier**. **Mandatory** rules (legal and accessibility, owned by compliance or accessibility) block from day one once the setup back-test shows they fire correctly. **Brand** rules: class A suggest and class B start advisory (shadow mode); either graduates to blocking when the lower bound of a 95% interval on reviewer agreement reaches 80% (e.g. 20 of 20, or 38 of 40). Class C rules are flagged by the rule linter on import and do not run until the owner adds a threshold or examples. Each class A rule names a **check type** (pattern, word list, count, contrast, presence) with parameters, imported from CSV; the **rule linter** flags rules whose text uses vague quantifiers ("sparingly", "where appropriate") without parameters. An AI compiler proposes rules from a PDF brand book (Round 3); a person approves each one.
B · Fact ledger	Facts as structured fields (value, unit, product, market, population, duration), with an owner, valid-from and valid-until dates, a confidentiality label, **approved wordings** per market and channel, and any **required disclaimer**. Imported from the team's existing sheet; code checks work from day one even with an empty ledger, and a new brand can approve its top 20 facts and 30 rules in about a day; at onboarding an AI drafts candidate facts from product pages and past approved content (Round 3). Every fact stays "unverified", and cannot back a claim or a fix, until its owner confirms it.
C · Check engine	Code checks, then AI judgement, then citation and support checks. It also checks for **required content** (disclaimers, risk warnings, paid-partnership labels) and produces a **coverage statement**. Tone rules are class B rules with good and bad examples from the brand book: the model compares each sentence with them, must cite the rule and quote the span, and starts advisory.
D · Explain and fix	A card per finding: the quote, the reason, the rule or fact with its source excerpt, how it was decided, and a fix. Claim fixes are **swaps from approved wording**; other fixes pass the five-part intent lock (§6.2).
E · Review workflow	Private pre-checks, statuses, risk routing, comments, decisions with reasons, version diffs and an exportable, hash-chained audit log (Round 2); a second approver and the publish-gate API (Round 3).
F · Visual and accessibility	Only measurable checks: logo variant and retired lockups, distortion, clear space, minimum size, contrast, palette on graphic elements (never inside photos), resolution, text inside images, alt-text presence, heading order and link text. Accessibility is a floor the dial cannot weaken.
G · Insights and guideline health	Recurring issues by rule, channel and origin; override and disagreement rates per rule; rule maturity; dormant and conflicting rules; reviewer feedback that contradicts the rulebook; recall tasks when facts expire.
H · Trust layer	Roles, confidentiality labels, code-only matching of confidential facts with matched draft spans masked before any AI call, PII redaction, an exportable **egress ledger** of every external call, and a plain disclosure of every AI service used.

4.4 A worked example: one post, start to finish

An agency drafts an Instagram post for India with an AI writing tool and submits it with a banner. Pinned intent: *clicks to the product page; key message "visible results on dark spots"; tone energetic.* This is the **brand reviewer's view**; the agency's own pre-check withholds the embargo finding (§4.5).

Plumbline · Aurelle Skin · "Vitamin C launch post" · v1 · brand reviewer view — Origin: Agency + AI tool · Instagram · India

Meet the NEW[5] Aurelle Vitamin-C serum[6]!!! 100% natural, eco-friendly[2] and clinically proven to fade dark spots by 60% in just 2 weeks[1]. Dermatologist recommended[3]. Launching with our Mumbai flagship on 15 Nov[4]. Click here[7] to shop!



8	BLOCKING	Mandatory accessibility: banner text #B8B2A7 on #FBF6EC is 1.96:1, below even the 3:1 minimum for large text; brand Ink on Cream is 13.24:1
9	MAJOR	Graphic block #FF3EA5 is ΔE_{00} 42.7 from the nearest brand colour (tolerance 10); logo stretched 12% (VI-04)
10	BLOCKING	Mandatory accessibility: no alt text. Plumbline asks: "What must someone who can't see this image learn from it?" It writes no alt text itself.
11	BLOCKING	Banner text "GLOW IN 2 WEEKS" (read by OCR) contradicts FACT-014 (8 weeks). Approved wording AW-014c: "Visible results in 8 weeks*".

Intent lock on the suggested version

PASS	1 · Call to action and product link kept
PASS	2 · Pinned phrase "dark spots" kept
PASS	3 · No claim dropped silently: 3 swapped, 1 flagged
PASS	4 · Within Instagram caption limits
PASS	5 · Edits stay inside flagged sentences
CONFLICT	Not one of the five checks: pinned tone "energetic" conflicts with brand rule TN-02; shown to the creator, who chose TN-02

BLOCKING · CLAIM · CONTRADICTED

AI EXTRACT + CODE COMPARE

"*clinically proven to fade dark spots by 60% in just 2 weeks*"

FACT-014: 32% reduction in visible dark spots, 8 weeks, in-house study, 48 participants · owner: Compliance · valid to Mar 2027

Approved wording AW-014b: "shown to visibly reduce dark spots by 32% in 8 weeks*" + required footnote DL-02

Swap in | Other approved wording | Ask compliance

BLOCKING · CLAIM · REGULATED PHRASE

MANDATORY RULE · CODE

"*100% natural, eco-friendly*"

CL-03 "Environmental claims must be specific and substantiated" · mandatory tier, owned by Compliance · basis: CCPA greenwashing guidelines 2024

Approved wording AW-009a: "made with 94% naturally derived ingredients" (FACT-009, ISO 16128)

BLOCKING · CONFIDENTIAL · EMBARGO

CODE ONLY · SPAN MASKED BEFORE AI

"*Launching with our Mumbai flagship on 15 Nov*"

Matches FACT-031 (embargoed to 1 Dec 2026) · details for Brand and Compliance only; the agency sees "restricted information"

Fix: remove the sentence

3	BLOCKING	"Dermatologist recommended": endorsement claim with no matching fact (Unsupported). No approved wording exists, so no fix is offered.
5–7	MINOR	TN-02 capitals and "!!!"; TM-007 product name is "Vitamin C Brightening Serum"; CTA-02 calls to action name the product, not "Click here"

7 blocking · 1 major · 3 minor · code findings in 3 s, full report in 38 s · rulebook v1.3 · models logged · confidential spans masked before AI
Not checked: claims implied by imagery, photography style, the screen-reader experience, and anything said in video or voice-over. These need a person.

Suggested version (intent kept, no invented claims)

Meet the new Aurelle Vitamin C Brightening Serum. Shown to visibly reduce dark spots by 32% in 8 weeks.* Made with 94% naturally derived ingredients. Shop the Vitamin C Brightening Serum.

*In-house study, 48 participants, 2026 (required footnote DL-02). Claims use approved wordings AW-014b and AW-009a.

Banner: "Visible results in 8 weeks*" with its footnote (AW-014c), in Ink on Cream, logo restored.

What people still decide

The agency accepts or edits each fix. Compliance chooses the claim wording and confirms the footnote. The design reviewer approves the corrected banner; the accessibility approver (in a small team, the brand owner) confirms the contrast fix and the creator's alt text. History: v1 submitted with 11 findings (the agency's pre-check showed 10); v2 accepted 8 fixes; v3 approved, hash 9f3c…e1. Any later edit re-opens the check.

Aurelle Skin and all its facts are fictional demo data. Contrast and colour distance were computed with the WCAG 2 formula and CIEDE2000; the timing and the 12% stretch are illustrative. Rule IDs belong to the fictional brand book. Contrast and alt-text checks (8, 10) ship in Round 2; palette, logo and text-inside-image checks (9, 11) in Round 3.

4.5 Review and approval workflow

Statuses: PRIVATE PRE-CHECK → SUBMITTED → IN REVIEW ⇄ CHANGES REQUESTED → APPROVED → PUBLISHED (or WITHDRAWN / RECALLED). Reviewers never see pre-checks. Every transition is written to the audit log with who, when and why. "Reviewer" includes the design and accessibility approvers, each deciding only its own category.

ACTION	CREATOR	EXTERNAL CREATOR	REVIEWER	COMPLIANCE	BRAND OWNER	AUDITOR
Run private pre-checks; see own findings	●	●	●	●	●	–
See confidential sources and evidence	◐ by label	–	●	●	●	●
Dismiss a minor finding (reason required)	●	●	●	●	●	–
Decide a major finding	–	–	●	●	●	–
Decide a blocking claim or confidentiality finding	–	–	–	● + 2nd approver	–	–
Approve content	–	–	●	● claims	●	–
Edit rules (each function edits only its own)	–	–	–	● regulatory	● brand	–
View insights, guideline health and the audit log	◐ own	◐ own work	●	●	●	●

- **Small teams:** the plan never needs more people than the team has. In a team of two or three, the brand owner can hold the compliance or accessibility role, but a second approval must always come from a different person than the first decision; the audit log records who held which.
- **Routing by risk:** any claim or confidentiality finding adds Compliance as a required approver; visual findings add the design reviewer; accessibility findings can only be decided by a named accessibility approver. Low-risk content with no open findings goes to one reviewer.
- **Confidentiality for agencies:** an external pre-check never reveals whether text matches a confidential fact, so it cannot be used to probe for secrets; the finding appears for the brand team on submission, and the agency's coverage statement says "confidential checks run on submission". No-login links (Round 3) are per brief, expire after 7 days and are rate-limited.
- **Not the regulatory record:** Plumbline's approval is the workflow record. Regulated firms export it, with the evidence and audit trail, to their own promotions register; it does not replace formal sign-off.
- **Change tracking:** every version is stored with a diff, comments carry across versions, and each decision is recorded with its reason. Any edit after approval re-opens the check.

4.6 Insights, guideline health and claim recall

Insights (team leads)

- Recurring issues by rule, channel and **origin** (employee, agency, AI tool), from submitted content only
- First-pass clean rate, review rounds, time to approval
- Claim coverage, and facts expiring in the next 30 days
- **Claim recall:** when a fact changes or expires, every published or scheduled asset using it gets an update-or-withdraw task for its owner

Guideline health (brand owner)

- **Override rate** and **reviewer disagreement** per rule (v1 at Round 2)
- **Rule maturity:** brand rules graduate to blocking when the 95% lower bound of reviewer agreement reaches 80%. In Round 2 the decisions come from testers labelling a back-test of 60 past posts, so only frequent rules will graduate that early.
- Dormant and conflicting rules; reviewer feedback that contradicts the rulebook ("moving goalposts")
- A drafted clearer rule from real accepted and dismissed cases, for the owner to approve

> **Patterns, not people.** Pre-checks never feed insights. No breakdown is shown for groups smaller than three people, there are no individual leaderboards, and an agency sees the same views about its own work that the client sees, including how often the client's reviewers contradicted their own rulebook.

5 Planned features, formats and scope

FEATURE	PRIORITY	MILESTONE	NOTES
Auth, organisations, roles, row-level security; demo data and golden set	MUST	M1	Foundation
Rule editor and import (Markdown, CSV); rule tiers; a rule linter that flags vague (class C) rules	MUST	M2	Improvement 3 groundwork
Fact ledger: structured facts, approved wordings, disclaimers, CSV import	MUST	M2	Improvements 1 and 2
Text checks (paste incl. reel scripts and captions, DOCX, PDF): rules, required content, claims, AI judgement	MUST	M2	Improvement 1
Citation and support checks; coverage statement; evaluation harness and baseline	MUST	M2	Improvement 1 evidence
Finding cards; approved-wording swaps; intent lock; accept / edit / dismiss; private pre-check	MUST	M3	Improvement 2
Accessibility floor on images: text contrast, alt-text presence	MUST	M3	Accessibility in Round 2
Review workflow: statuses, routing, comments, decisions, audit log and export; override and disagreement table	MUST	M4	Human control; Improvement 3 v1
Slack results; no-login check link; publish-gate API; second approver; DOCX redlines; HTML input	SHOULD	M5	Where creators work
Palette on graphics (photo regions separated), logo variant, retired assets, distortion, clear space; claims in image text (OCR, vision fallback; low confidence reported as not checked); heading order	SHOULD	M6	Deeper visual checks
Rule maturity automation; drafted clarifications; claim recall; AI fact drafting; Sheets sync; rulebook compiler	SHOULD	M7	Improvement 3 full
URL post-publication watch; reel audio transcripts; Hindi and Hinglish; EU regime	COULD	Roadmap	Influencer and video content
Canva, Figma and Google Docs add-ons; self-hosted model option	COULD	Roadmap	Meet creators inside their tools
Auto-publishing · AI-written claims or alt text · individual scores · training on customer content	WON'T	Never	Deliberate anti-goals

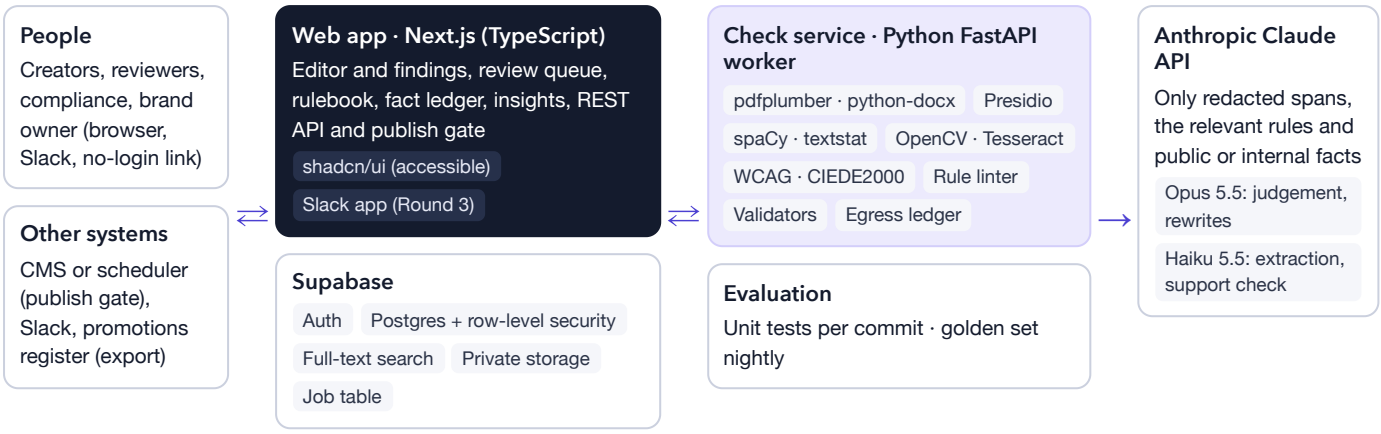
Formats, in build order: 1 PLAIN TEXT → 2 DOCX → 3 PDF → 4 PNG / JPG (M3, M6) → 5 HTML (M5) → ROADMAP: REEL TRANSCRIPTS, PPTX. Everything is parsed into one document model (text blocks with offsets, images with positions, links, headings, alt text), so every check works on every format. In Round 2, "Accept" produces a corrected text block and a change list; in Round 3, DOCX gets real tracked changes written as w:ins and w:del elements in the document XML (python-docx has no tracked-changes API).

Scope guardrails: one vertical and regime (Indian D2C personal care) and English (Indian, British and American varieties) for the build; one demo organisation with realistic data. For influencer reels, Plumbline checks scripts, captions and cover frames; spoken audio is not checked until transcripts arrive (roadmap), and the coverage statement says so.

TECHNICAL PLAN

6 Architecture, stack, integrations and data

6.1 Architecture



6.2 The check pipeline

#	STAGE	RUNS AS	WHAT HAPPENS
1	Ingest	CODE	Parse to the document model; extract images, OCR text and boxes, links, headings and alt text; separate photo regions from flat graphic regions by colour count per tile (a heuristic, used from M6).
2	Redact and gate	CODE	Presidio redacts people's names, contact details and ID numbers only, with an allow-list of brand and product names; dates, places and numbers stay because claims depend on them. Draft spans that match confidential or embargoed facts (exact, fuzzy, alias and date matching) are masked; those facts never enter a prompt. Paraphrases the matcher misses are the residual risk, so its catch rate is measured.
3	Rule engine	CODE	Class A rules: word and phrase lists, regulated-phrase lexicon, required content present, formatting, readability, link text, heading order, alt-text presence, contrast, palette, logo, embargo matching, numeric claim comparison.
4	Claims and retrieval	CODE + AI	Claims found by regex detectors (numbers, superlatives, regulated phrases) and by Haiku extraction. Each is matched to facts on structured fields (product, market, unit) and Postgres full-text search. A match below a relevance threshold becomes "needs human eyes", never a guess.
5	AI judgement	AI	Opus 5.5 with structured output judges class B rules, including tone against the brand book's examples, and whether claims are supported (public and internal facts only). Calls run per section, in parallel. The approved class B rulebook sits in a cached prompt prefix; facts are added per claim. Content is delimited and treated as data, never as instructions.
6	Validate	CODE + AI	Drop any finding whose rule or fact ID doesn't exist or whose quote isn't in the content. Then a second model call (Haiku) checks that the cited source actually supports the finding; disagreement becomes "needs human eyes". Dropped findings are logged and measured.
7	Fix and intent lock	CODE + AI	Claims: only approved wordings for that market and channel are offered. Other rewrites must pass five checks: **(1)** call to action kept; **(2)** pinned phrases kept in meaning; **(3)** no claim silently dropped; **(4)** channel limits met; **(5)** edits stay in the flagged sentence. Every fix is re-run through all rules and must add no new findings.
8	Report	CODE	Findings plus a **coverage statement** (rules run, rules in shadow mode, what is out of scope), saved with rulebook, model and prompt versions, inputs and outputs, Code findings appear within seconds; AI findings stream in as each section finishes. Runs are **traceable, not reproducible**: model outputs can vary between runs, so each run's record is kept.

Finding record (simplified)

```
{ "id": "F-0193", "category": "claim",
  "severity": "blocking", "status": "contradicted",
  "quote": "clinically proven to fade…",
  "span": { "start": 84, "end": 139 },
  "basis": { "type": "fact", "id": "FACT-014",
             "excerpt": "32% … after 8 weeks" },
  "fix": { "type": "approved_wording",
           "id": "AW-014b", "disclaimer": "DL-02" },
  "support_check": "agree",
  "method": { "by": "ai+code",
              "model": "claude-haiku-5-5" },
  "run": { "rulebook": "v1.3", "prompt": "claims@7" }
}
```

Claim verification logic

```
# each claim: regex detectors + AI extraction
if regulated_phrase(claim): flag("needs evidence")
fact, score = best_match(claim, ledger)
if fact is None:
    status = "unsupported"
elif score < MATCH_THRESHOLD or not fact.verified:
    status = "needs human eyes"
elif fact.valid_until < today:
    status = "outdated"          # expired source
elif fact.code_only or claim.has_number:
    status = compare(claim, fact)     # code
else:
    status = ai_entails(claim, fact)  # AI
fixes = approved_wordings(fact, market, channel) if
fact else []
emit(status, basis=fact, quote=claim.span,
     fix=fixes or None)    # never generated
```

6.3 Technology stack, and why

LAYER	CHOICE	WHY
Web app	Next.js (React, TypeScript), Tailwind CSS, shadcn/ui on Radix primitives; Slack app via Bolt	One codebase for UI, API routes and the Slack surface; accessible components
Data and auth	Supabase: Postgres with row-level security, Auth, Storage and full-text search (no vector store needed in Round 2)	Tenant isolation enforced in the database; one managed service for a solo builder
Check service	Python 3.12 + FastAPI worker, with jobs in a Postgres table	Best document, vision and language libraries; no extra queue to run
Check libraries	pdfplumber (MIT licence, chosen over AGPL-licensed PyMuPDF), python-docx (parsing); spaCy small model, textstat (language); OpenCV, Tesseract OCR, k-means palette + CIEDE2000, WCAG contrast (visual)	Deterministic and explainable; exact measurements for class A rules
AI models	Anthropic Claude: **Opus 5.5** for judgement and rewrites; **Haiku 5.5** for claim extraction and support checks; structured outputs and prompt caching. Sonnet 5.5 is the fallback if evaluation shows equal quality at lower cost.	Schema output feeds the validators; vision support; caching the rulebook cuts cost and delay
Privacy tooling	Microsoft Presidio (PII redaction, with an allow-list of brand and product names); local ONNX embeddings only if keyword matching proves too weak (Round 3)	Personal data removed before any external call, without redacting the brand's own names
Hosting and cost	Vercel free tier; Supabase Pro (US$25 a month) during judging so the demo never pauses; worker on a small paid instance with 2 GB RAM (about US$10–25 a month)	Worker footprint under 1.5 GB; ≈ US$0.10 per 1,000-word check including thinking tokens (§6.7); API budget ≈ US$190

6.4 Data model

`Organisation · User · Membership(role, function) · GuidelineVersion · Rule(id, category, class, check_type, params, severity, examples, source_ref, owner_function, tier, maturity) · Fact(value, unit, product, market, population, duration, owner, valid_from, valid_until, confidentiality, status) · ApprovedWording(fact, market, channel, text, disclaimer) · Term · Asset · Content · Version(canonical_hash, origin, intent, private) · CheckRun(rulebook, model, prompt, input_hash, coverage) · Finding · Decision(kind, reason, actor) · Comment · Approval · FactUsage(version, fact) · EgressEvent · AuditEvent(prev_hash)`

6.5 Integrations, in build order

1. **Exports** (M4): audit log and decisions as CSV for a promotions register.
2. **Slack app and no-login check link** (M5) for creators and agencies.
3. **REST API and publish gate** (M5): where a scheduler or CMS can call it, unapproved versions are blocked. Where nothing can call it (Meta Ads Manager, influencers' phones), the approval is the record, and post-publication watch (roadmap) compares what went live. The version hash is computed over a canonical form so any caller can recompute it.
4. **Google Sheets fact sync** (M7).

6.6 Data sources

- **From the organisation:** brand book, fact sheet with approved wordings and disclaimers, terminology, asset kit, past approved content.
- **Public references:** WCAG 2.2 [19]; a regulated-phrase list compiled from the CCPA guidelines, the ASCI code and influencer guidelines, the Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954 and the Cosmetics Rules, 2020, and (roadmap) EU 2024/825 and the CMA Green Claims Code.
- **Evaluation:** the fictional Aurelle Skin brand with 150 seeded items and 60 past posts for the back-test (disclosed as synthetic); 60 public ASCI decisions (complaints upheld and not upheld) as an independently labelled set.

6.7 Reliability, evaluation and cost

The evaluation harness is a Must in M2. Deterministic unit tests run on every commit at no API cost; a 40-item smoke set runs nightly and the full set at each milestone, both through the Message Batches API at half price. The builder writes 100 synthetic development items with AI assistance; the **50 held-out items (about 120 seeded issues) are written and labelled by two recruited marketing practitioners** (agreement reported as κ), named with consent in the Round 2 report, and are scored only at the Round 2 and Round 3 report points, so they stay unseen. Each target is a point estimate reported with its count and Wilson 95% interval, and compared with a **single-prompt baseline** (brand book and content in one Opus 5.5 call). Public ASCI decisions test claim detection and risk ranking, not substantiation: claims are transcribed from ASCI's published complaint decisions, the rulebook is my regulated-phrase list and generic claim rules, and the fact ledger is empty. A hit is flagging the challenged claim (span overlap). I report recall on upheld complaints and how well upheld claims are ranked above not-upheld ones (AUC); with no evidence ledger, precision would be meaningless, so it is not reported. The labellers also check the citation sample.

MEASURE	ROUND 2	ROUND 3
Citation correctness (human-checked, n = 100)	≥ 90 / 100	≥ 95 / 100
Recall, seeded claim issues (held-out, n ≈ 120)	≥ 80%	≥ 90%
Recall / ranking (AUC), ASCI decisions (n = 60)	≥ 70% / ≥ 0.65	≥ 80% / ≥ 0.70
Precision per category, incl. tone (held-out)	≥ 75%	≥ 85%
Precision gain over single-prompt baseline	measured	≥ 10 pts
False alarms on clean items (n = 30)	≤ 1 / item	≤ 0.3 / item
True issues wrongly dropped by validators	≤ 10%	≤ 5%
Embargo matcher: paraphrases caught (n = 40)	≥ 70%	≥ 85%
Invented claims in a 200-fix audit	0	0
PII redaction recall: Indian names, phone, Aadhaar, PAN (n = 50)	≥ 90%	≥ 95%
Recall drop under injection ("ignore rules", hidden issues)	≤ 10 pts	≤ 5 pts
Code findings · full report p95 (1,000 words + 2 images)	< 5 s · < 60 s	< 5 s · < 40 s

CALLS PER 1,000-WORD CHECK	TOKENS IN / OUT	US$
Claim extraction (Haiku 5.5)	2,000 / 500	0.0005
Judgement (Opus 5.5): cached rulebook	6,000 cached	0.0012
Judgement (Opus 5.5): content and facts	3,000 / 1,500	0.0420
Judgement thinking (effort medium, the default)	~1,500 out	0.0300
Support check (Haiku 5.5)	3,000 / 300	0.0005
Rewrites for non-claim findings (Opus 5.5, effort low)	2,500 / 900 incl. thinking	0.0280
Intent check on rewrites (Haiku 5.5)	1,500 / 200	0.0003
Total per check (image checks are code only)		**≈ 0.10**

List prices [28]: Opus 5.5 US$4 / US$20 per million input / output tokens (cache reads US$0.20); Haiku 5.5 US$0.10 / US$0.50; Haiku's own thinking adds under US$0.001 a check; a cold cache write (≈ US$0.03) occurs once per cache window. **Budget:** ~500 development checks ≈ US$50; 8 full runs of 210 items at batch price ≈ US$85; nightly smoke runs ≈ US$30; baseline runs ≈ US$20; total ≈ US$190.

RESPONSIBLE DESIGN

7 Privacy, security and human oversight

Privacy and confidentiality

- Every fact and asset carries a label: Public, Internal, Confidential or Embargoed-until-date. **Confidential and embargoed facts are checked by code only** and never placed in a prompt; draft spans that match them are masked before any AI call, and I measure how often the matcher catches paraphrases. Paraphrases it misses could reach the AI, so sensitive launches can run in code-only mode.
- **Code-only mode:** organisations that cannot send content out run only the rule engine (nearly half the rules in the guide I audited) with zero external calls.
- PII is redacted before any AI call; only the needed spans and rules are sent; embeddings are computed locally.
- An exportable **egress ledger** records which text went to which model and region, and when.
- Designed for India's Digital Personal Data Protection Act, 2023: personal data is minimised and redacted, processed under a data processing agreement, and deleted on request.
- I use Anthropic's commercial API, whose terms exclude training on API inputs by default (re-confirmed at build time); data residency options are reviewed per customer.

Security

- Row-level security on every table, with automated cross-tenant access tests. For each job the app issues the worker a short-lived token scoped to that job's organisation, so row-level security still applies; the worker holds no service key.
- The audit log is insert-only (UPDATE and DELETE revoked) and hash-chained; the daily chain head is e-mailed (via a disclosed SMTP provider) to the brand owner and compliance as an external anchor, so later edits are detectable. It exports to CSV.
- Private storage with short-lived signed links; file type and size checks; uploaded files are parsed in a sandboxed process with no database credentials; secrets outside the repository.
- **Prompt injection:** content is delimited data; the model has no tool that can change state; outputs are schema-validated and citation-checked; I test both "approve me" and "hide this issue" attacks; only people approve. Cache keys include rulebook, model and prompt versions.

Human oversight

- **Autonomy dial** per category: one-click fix (mechanical only, still needs a click), suggest, or require a person. Claims and accessibility can never be set to one-click.
- Mandatory legal and accessibility rules block from day one; brand rules advise until they earn the right to block.
- Creators cannot dismiss blocking findings; compliance decides claim findings with a second approver; each function owns its own rules.
- Nothing publishes without a human approval bound to a version hash.

Transparency and explainability

- Each finding shows its rule or fact ID, the source excerpt, the reason and how it was decided (rule engine, or AI with model name and support-check result). A confidence number appears only after calibration on held-out data.
- Every report carries a **coverage statement** and never a pass badge, score or "compliant" label.
- A "What leaves this system" page lists every external service, what data it receives and why.
- Results are traceable: each run stores its rulebook, model and prompt versions, inputs and outputs.

Fairness, accessibility and people

- **Accessibility is a floor:** the brand owner cannot weaken it, contrast wins over palette, and only an accessibility approver can decide those findings.
- Plumbline never writes alt text or claims; it asks the creator.
- Tone rules must not penalise regional varieties of English; the evaluation set includes them.
- "Patterns, not people": private pre-checks, no breakdown for groups under three, symmetric views for agencies. The interface targets WCAG 2.2 AA.

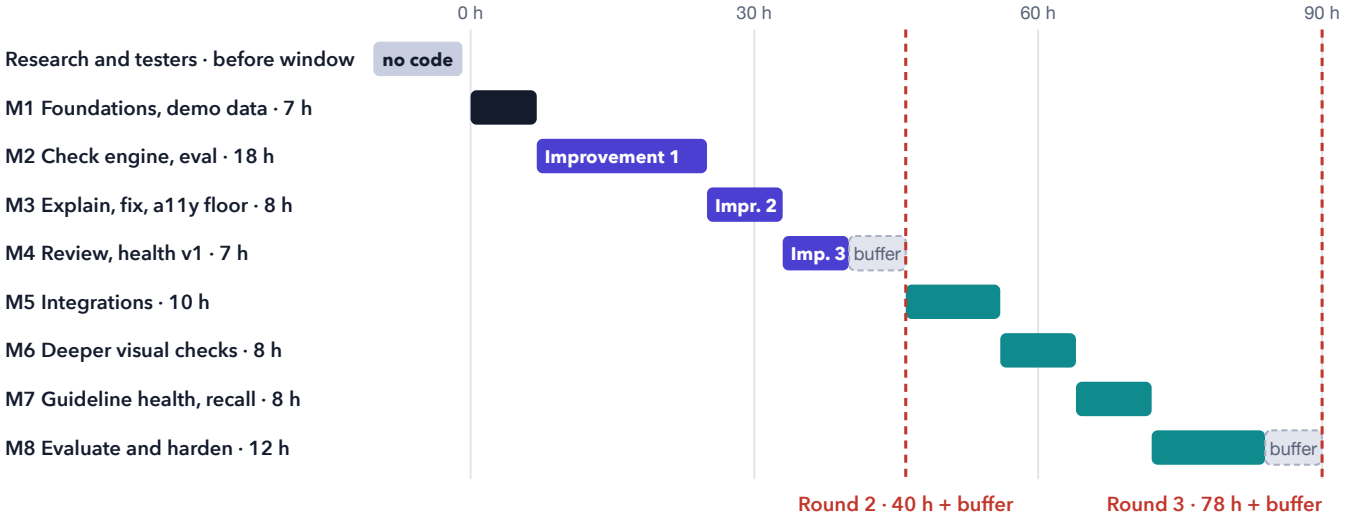
Limits I state up front

- Automated accessibility checks cover only part of WCAG; reports say so.
- AI judgement can be wrong. Citations, support checks and confidence make errors visible, and people decide.
- **Not legal advice** and not a regulatory sign-off; legal approval stays with qualified reviewers.
- Photography mood, implied claims in imagery and video are out of scope until proven measurable.

FEASIBILITY AND SCOPE

8 Development plan

The schedule is sized in hours of solo work with AI coding assistance: **40 engineering hours for the Round 2 core**, 78 engineering hours in total, **plus 12 hours of research and user testing and a 15% buffer (12 hours), 102 hours overall**. Nothing is coded before the window opens (rules 1 and 4); research, design and recruiting testers happen before it (rule 2). Hours convert to dates once the window is announced.



MILESTONE	HOURS	DELIVERABLE (EACH MUST BE DEMONSTRABLE, TESTED AND COMMITTED)
M1 Foundations and demo data	7	Repository, CI, schema with row-level security, auth and roles, upload; seeded Aurelle Skin organisation, rulebook, 40 facts, golden set.
M2 Check engine and eval harness	18	Document model; rule engine with tiers, required-content checks and the rule linter; rule import; fact ledger with approved wordings; claim matching; AI judgement including tone; citation and support checks; coverage statement; evaluation harness and baseline.
M3 Explain, fix, accessibility floor	8	Finding cards, approved-wording swaps, intent lock, accept / edit / dismiss, re-check, diffs, private pre-check; text contrast and alt-text presence on images.
M4 Review workflow, health v1	7	Statuses, routing, comments, decisions, hash-chained audit log and export, override and disagreement table, back-test of 60 past posts. **Round 2 demo:** the full loop for text, DOCX and PDF with the accessibility floor on images; v1 of all three improvements; counterbalanced tests with 6 users.
M5 Integrations	10	Slack results, no-login link (expiry, rate limits), publish-gate API with canonical hash, second approver, DOCX redlines, HTML input.
M6 Deeper visual checks	8	Palette on graphics, logo variant, retired assets, distortion, clear space; claims in image text (OCR with vision fallback); heading order.
M7 Guideline health and recall	8	Rule graduation automation, drafted clarifications, claim recall tasks, AI fact drafting, Sheets sync, rulebook compiler.
M8 Evaluate and harden	12	Full evaluation with intervals, injection and tenancy tests, confidence calibration, performance, fixes from user tests, documentation, demo video. **Round 3.**

Cut line. If a checkpoint slips, items move to the roadmap in this order: rulebook compiler → Sheets sync → AI fact drafting → logo checks → drafted clarifications (the guideline-health metrics stay) → DOCX redlines. Inside the 40-hour core, a working end-to-end skeleton (paste text → rule engine → findings → approve) exists by hour 12. If the core itself slips, cut in this order: PDF input → AI tone rules → private pre-check → contrast on images. Never cut: claim checks with citations, the coverage statement, approved-wording swaps, the intent lock, one-reviewer approval with the audit log, alt-text presence, and the override table (fed by user-test decisions if the back-test is cut).

Scope for each possible window length

IF THE WINDOW IS	ROUND 2 SHIPS	ROUND 3 ADDS	MOVES TO ROADMAP
48 hours (~32 h of real work)	The core after the in-core cut order (~32 h): v1 of all three improvements and alt-text checks ship	Back-test; contrast on images; evaluation report	M5–M7
About 1 week (~60 h)	M1–M4 (40 h) plus buffer (6 h)	M8 (12 h), total 58 h	M5–M7
2 weeks or more	M1–M4 plus user tests	M5–M8 in full	Add-ons, URL watch, transcripts

Risks and mitigations

RISK	LIKELIHOOD	IMPACT	MITIGATION
AI flags the wrong things	Medium	High	Code first; citation and support checks; shadow mode until rules earn trust; nightly evaluation
A solo builder runs out of time	High	High	40-hour core, 15% buffer, cut line and window table; daily commits; disclosed AI coding assistance
The fact ledger is empty or stale	High	Medium	Import from existing sheets; AI-drafted candidate facts; "unverified" until the owner confirms; owner nudges; sync from the source; freshness as a KPI
Confidential data exposure	Medium	High	Code-only matching; masking before AI with a measured catch rate; code-only mode for sensitive launches; no signal to agencies; egress ledger; tenancy tests
Too few real users for testing	Medium	Medium	Recruit 5 to 8 testers and the held-out labeller before the window; structured task-based sessions

FEASIBILITY AND IMPACT

9 Role allocation

Team AtoZ has one member, who covers every role. Each role has a fixed share of the 102-hour budget, so every piece of work has an owner and a limit.

ROLE	HOURS	RESPONSIBILITIES
Product and research	12	Interviews and survey before the window, user-test sessions, scope and cut-line decisions
Full-stack engineering	32	Next.js app, Supabase schema and row-level security, review workflow, integrations
AI and checks engineering	30	Rule engine, fact ledger, claim matching, prompts, validators, golden set and evaluation
Design, QA and documentation	16	Interface and accessibility, demo data, test plans, README and architecture notes, demo video
Buffer	12	15% of engineering hours, held back for slips

Disclosure (rule 7). Claude (Anthropic) was used for research synthesis, drafting, the synthetic persona panel, the blind second rating, and AI reviewers that critiqued drafts of this plan. It will be used as a coding assistant during the build; all AI-generated code will be reviewed, tested and logged in `AI_USAGE.md`. No code exists yet. Starter templates (e.g. Next.js, shadcn/ui) will be declared in the README.

10 Expected impact

For people

- **Creators and agencies** get specific, private feedback in seconds, and fewer rejected drafts.
- **Reviewers** see mechanical issues cleared first, and claims arrive with evidence and approved wording.
- **Brand owners** see guidelines improve with use, instead of a static PDF.

For the organisation

- An evidence trail for every published claim, and recall when a fact changes, as CCPA penalties reach ₹10 lakh, or ₹50 lakh for repeats, for brands and endorsers.
- Fewer accessibility failures, and a consistent brand across employees, agencies and AI tools.

MEASURED IN THE PRODUCT FROM M4	WHY IT MATTERS
First-pass clean rate; review rounds per asset; time to approval	The bottleneck that grew from three rounds to five [10]
Claim coverage; live assets using expired facts	Evidence behind every claim; nothing outdated left live
Rule override rate; share of rules matured to blocking	Are the guidelines getting clearer? (improvement 3)
Repeat-issue rate over time, by group	Does feedback actually teach?

Baseline test (Round 2): two matched sets of 20 seeded posts (A and B); half the testers review A with the brand book PDF and B with Plumbline, the other half the reverse, so order and familiarity cancel out. Targets to test: ≥ 30% more seeded issues caught and ≥ 40% less time per post.

> **Illustrative estimate, to be validated.** If a brand team approves 200 assets a month and a review round takes 20 reviewer-minutes, removing one round per asset saves 200 × 20 min ≈ **67 reviewer-hours a month**, about 0.4 of a full-time reviewer. Interviews will replace these assumptions.

Value case and price (hypotheses, tested by H5): the head of brand or CMO buys, with compliance as co-sponsor. The value is avoided rework and withdrawn campaigns, an evidence trail for regulators, and reviewer time. Price per organisation, not per seat: a free code-only tier, then a first band to test of ₹40,000–75,000 a month, about 2–4 times the AI cost of 2,000 checks (≈ US$200, roughly ₹17,000). **Beyond the event:** creator-tool add-ons, Hindi and Hinglish, reel transcripts and post-publication watch, self-hosted models, SSO and SCIM.

APPENDICES

Appendix A · Sources

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Appendix B · Style-guide audit: method and all 114 rules

Scope: five sections of the Mailchimp Content Style Guide [15], retrieved October 2026, paraphrased with credit under CC BY-NC 4.0. **Unit:** one directive statement; descriptive sentences excluded, duplicates counted once, mixed sentences split. **R1:** the author, with AI assistance. **R2:** an AI model given only the rule list and class definitions, blind to R1.

Shaded rows: the raters disagreed (20 of 114).

ID	R1	R2	RULE (PARAPHRASED)
VT01	C	C	Be plainspoken; avoid fluffy metaphors and cheap emotional plays
VT02	C	C	Be genuine: familiar warm and accessible
VT03	B	C	Demystify B2B-speak and educate
VT04	C	C	Humour is dry and subtle; weird but not inappropriate
VT05	C	C	Tone usually informal; clarity beats entertainment
VT06	C	C	Adjust tone to the reader's emotional state
VT07	C	C	Be funny only when appropriate and natural
VT08	B	B	Use active voice (exception when emphasising the action)
VT09	B	C	Avoid slang and jargon; write plain English
VT10	B	C	Write positively rather than negatively
VT11	B	B	Do not write in the mascot's voice
GM01	B	C	Group related ideas; use descriptive headers
GM02	B	B	Lead with the main point
GM03	C	C	Be concise: short words and sentences
GM04	C	C	Be specific; avoid vague language
GM05	B	C	Spell out unfamiliar acronyms on first use
GM06	A	A	Email addresses and URLs in lowercase
GM07	A	A	Never capitalise website/internet/online/email mid-sentence
GM08	C	C	Use emoji infrequently and deliberately
GM09	B	A	Numerals except at sentence start (idioms excepted)
GM10	A	A	Numbers over three digits get commas
GM11	B	B	Write big numbers in full unless space is tight
GM12	A	A	Spell out day and month in dates
GM13	A	A	Spell out fractions
GM14	A	A	Use the % symbol not the word percent
GM15	A	A	Use a hyphen for ranges
GM16	A	A	Currency symbol before amount; cents only if non-zero
GM17	A	A	Phone numbers with dashes and country code
GM18	A	A	Temperature with degree symbol and F
GM19	A	A	Time as numerals with am/pm; no :00
GM20	B	B	Specify time zones for schedulable events
GM21	A	A	US time zone abbreviations; international zones spelled out
GM22	A	A	Abbreviate decades within the last 100 years
GM23	A	B	Possessive apostrophe rules
GM24	B	A	Use a colon to offset a list
GM25	A	B	Use the serial (Oxford) comma
GM26	C	C	Other commas: use common sense
GM27	A	A	Hyphen without spaces for compounds and ranges
GM28	A	A	True em dash without spaces
GM29	C	C	Ellipses sparingly; not for drama
GM30	A	A	Periods inside quotes; parenthesis placement
GM31	A	A	Single space between sentences
GM32	A	B	Question mark placement with quotes
GM33	C	C	Exclamation points sparingly
GM34	A	A	Never more than one exclamation point at a time
GM35	B	B	No exclamation points in failure messages or alerts
GM36	A	B	Quotation mark usage for words titles and quotes
GM37	A	A	Single quotes inside double quotes
GM38	C	C	Go easy on semicolons
GM39	A	A	No ampersands except in brand names
GM40	A	A	File extensions uppercase; filenames lowercase
GM41	B	B	Singular they when gender unknown; never one as pronoun
GM42	B	A	Quotes attributed in present tense
GM43	B	B	Full name on first mention then first name
GM44	B	B	Capitalise team names but not the word team
GM45	B	B	Capitalise job titles only for a specific role
GM46	A	A	Do not call people ninja rockstar or wizard
GM47	B	A	School full name first then abbreviation
GM48	A	A	Spell out city and state names; AP city list
GM49	A	A	United States on first mention then US
GM50	B	A	Capitalise website names; no italics
GM51	A	A	Avoid spelled-out URLs; drop http://www
GM52	B	B	Legal entity name only in legal documents
GM53	A	A	Exact brand-name capitalisation
GM54	B	B	Refer to the company as we not it
GM55	A	A	Capitalise branded terms and plan names only
GM56	A	A	Honour other companies' own spellings
GM57	B	B	Refer to other companies as it not they
GM58	B	C	Define technical terms when used
GM59	B	B	Italics for long-work titles and UI labels
GM60	A	A	No underline; no mixed bold/italic/caps
GM61	A	A	Left-align text
WP01	A	A	Do not capitalise audience mid-sentence
WP02	B	B	Audiences and contacts are they not it
WP03	A	B	Always use the lookalike qualifier for that feature
WP04	B	B	Mention age only if relevant
WP05	A	A	No age descriptors like young old elderly
WP06	A	A	Avoid disability idioms
WP07	B	B	Mention disability only if relevant
WP08	A	A	Never suffer victim or handicapped
WP09	A	A	Do not call groups guys or women girls
WP10	A	A	Prefer gender-neutral job terms
WP11	B	A	LGBT terms as modifiers not nouns
WP12	B	A	Avoid lifestyle/preference/homosexual for LGBT people
WP13	B	B	Avoid same-sex marriage unless relevant
WP14	B	B	Use a person's communicated pronouns
WP15	B	B	Deaf and hard of hearing usage
WP16	A	A	No hyphen in dual heritage terms
WP17	B	B	Mention medical condition only if relevant
WP18	B	B	Never assume or label mental conditions
WP19	A	A	Capitalise culture and ethnicity names
WP20	B	B	Blind and low vision usage
WA01	B	B	Avoid directional language
WA02	A	A	Headings nested and consecutive
WA03	B	B	Most important information first
WA04	A	A	Use true lists not line breaks
WA05	A	A	Label form inputs and required fields
WA06	C	C	The shorter the form the better
WA07	A	A	Descriptive links; avoid click here
WA08	B	C	Plain language and short sentences
WA09	A	A	Alt text on all images
WA10	B	B	Alt text content matches image purpose
WA11	A	A	Captions or transcripts for all video
WA12	C	A	Aim for high contrast between text and background
WA13	B	B	Images are never the only carrier of meaning
SM01	A	A	Platform length limits for social copy
SM02	A	A	No 4 for for or u for you
SM03	C	C	Avoid excessive exclamation points
SM04	B	B	Do not publicly tag subjects with we-wrote-about-you
SM05	A	A	Never ask for retweets or likes
SM06	C	C	Hashtags rarely and deliberately
SM07	B	B	No trending hashtags to promote the brand
SM08	B	B	No comment on unrelated trending topics
SM09	B	B	Pause scheduled posts during major breaking news

Appendix C · Interview guide and survey

Interview guide (30 minutes, semi-structured)

1. Walk me through the last piece of content you approved or had sent back. What happened, step by step?
2. What did you have to check by hand? Which of those checks felt mechanical?
3. Tell me about a time something wrong or off-brand got published. How was it caught?
4. Where does the evidence for product claims live today? Who owns it?
5. Which brand rules do people interpret differently? Can you give an example?
6. How do agencies and AI tools fit into your content process today?
7. What would you never let software decide on its own?
8. What data could not leave your company for a tool like this?

Survey (10 questions, about 4 minutes)

1. Your role, and the size of your organisation
2. How many people or agencies create content your team reviews?
3. Average review rounds before approval (1, 2, 3, 4, 5+)
4. Share of review time spent on mechanical issues (0–100%)
5. How often off-brand or incorrect content gets published
6. Where claim evidence is kept (system, spreadsheet, email, nowhere)
7. Do you use AI tools to draft content? Which?
8. Preferred feedback: a list of issues / a suggested fix / both
9. Rank: speed, accuracy, explanation, control, privacy
10. May I contact you for a 30-minute test session?

Appendix D · Disclosures and glossary

ITEM	DISCLOSURE
AI assistance	Claude (Anthropic) assisted with web research, synthesis and drafting; with the first-pass audit classification; as the blind second rater; as the eight simulated personas in §3.4; and as reviewers that critiqued drafts against the published rubric. All sources are listed in Appendix A so each figure can be verified.
Synthetic panel	The §3.4 personas are fictional and AI-simulated. Their transcripts are kept, labelled as such, and used only as design critique; no figure from them appears in this plan.
Planned external services	Anthropic Claude API (redacted spans, rules, public and internal facts); Supabase; an SMTP e-mail provider (notifications and the daily audit digest); Vercel and a worker host; Slack. Each will be listed in the product's "What leaves this system" page.
Testers and labellers	The held-out labeller and user testers are recruited before the window, credited by name with consent in the Round 2 report, and are not team members.
Data and reused material	All demo data is fictional. Open-source libraries named in §6.3 under their licences; the Mailchimp guide (CC BY-NC 4.0) for the audit only. No pre-existing code. "Plumbline" is a working name chosen for this event.

TERM	MEANING IN THIS PLAN
Class A / B / C rule	Code-checkable / needs language or visual understanding / ambiguous as written (§3.2); class C rules do not run until clarified
Mandatory and brand tiers	Legal and accessibility rules that block from day one / brand rules that start advisory
Shadow mode (advisory)	A brand rule's flags are shown as advice and logged, but cannot block, until the 95% lower bound of reviewer agreement reaches 80%
Approved wording	Pre-approved phrasings of a fact for a market and channel, with any required disclaimer; the only text Plumbline may offer as a claim fix
Coverage statement	The part of every report listing rules run, rules in shadow mode, and what was not checked
Intent lock	The five checks every non-claim rewrite must pass (§6.2, stage 7)
Claim recall	Update-or-withdraw tasks raised for every asset that uses a fact once it changes or expires
Egress ledger	An exportable log of which text went to which external model, where and when

Appendix E · Synthetic panel (AI-simulated design critique)

Eight fictional personas, simulated by Claude, each answered the interview guide (Appendix C) before reading a one-page concept brief, then reviewed the concept and made a pilot decision. Transcripts are kept and labelled AI-simulated. This was used only to find design gaps before real interviews: nothing from it is used as evidence of demand, and every design change it prompted is tested as H6 in §3.5.

SIMULATED PERSONA	THEIR MAIN CONDITION	WHAT I CHANGED
Compliance manager, consumer health	"Silence is not clearance"; never invent claim wording; back-test before go-live	Coverage statement on every report; approved-wording swaps; shadow mode on past content
Content lead, regulated fintech	Check that required wording is present; Plumbline approval is not the regulatory sign-off	Required-content checks; record exports to the firm's promotions register; rules owned by function
Head of brand, D2C skincare	Influencer content skips the CMS; claims ad-libbed after approval	No-login check link; claim recall and post-publication watch (reel transcripts on the roadmap)
Security buyer, private bank	Confidential facts never reach an AI; a synced fact source, not a stale copy	Confidential facts checked by code only; egress ledger; fact sync from the source sheet
Agency creative director	A private pre-check; no one-sided "agency" dashboard	Pre-checks private; agencies see the same views as the client, including reviewers' "moving goalposts"
Brand design lead, retail	Measurable visual checks only; no font uploads; exceptions that stick	Visual scope narrowed; palette checked on graphics only, never inside photos; sticky exceptions
Accessibility consultant	No pass badge, no AI-written alt text, accessibility never weakened	Accessibility floor; alt-text adequacy asked as a question; "not checked" list
Junior social creator	Under a minute per post; works from Slack; not a scoreboard	Optional intent fields remembered per campaign; results in Slack; personal stats private